UNIVERSALLY BOUNDED FIBRES DESCEND ALONG SURJECTIVE FLAT MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 397-414, immediately following the statement at 383-395.

Editorial note. One cross-chapter pointer (Morphisms) is recorded in assembly.external_references.

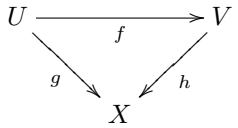
Allowed standard prerequisites (editorial).

- Definition of universally bounded fibres. Definition definition-universally-bounded of this chapter, cited at proof lines 397-399. Fixes what n bounds: degrees of Spec(k) x_X U over Spec(k) for all field points.
- Degree boundedness descends along flat surjective morphisms of schemes. Morphisms, Lemma morphisms-lemma-universally-bounded-permanence, cited at proof lines 411-413. The imported scheme-side permanence: if the source of a flat surjective morphism has degree <= n over k, so does the target.

Source reference. Original definition 03JL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 277–277. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03JB: focused-seeds/geometry/S9-decent-universally-bounded-permanence/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Consider a commutative diagram*



where U and V are schemes. If g has universally bounded fibres, and f is surjective and flat, then also h has universally bounded fibres.

Proof. Assume g has universally bounded fibres, and f is surjective and flat. Say $n \geq 0$ is an integer which bounds the degrees of the schemes $\mathrm{Spec}(k) \times_X U$ as in Definition 03JL. We claim n also works for h. Let $\mathrm{Spec}(k) \to X$ be a morphism from the spectrum of a field to X. Consider the morphism of schemes

$$\mathrm{Spec}(k) \times_X V \longrightarrow \mathrm{Spec}(k) \times_X U$$

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is flat and surjective. By assumption the scheme on the left is finite of degree $\leq n$ over $\mathrm{Spec}(k)$. It follows from Morphisms, Lemma 03JB that the degree of the scheme on the right is also bounded by n as desired. $\square$